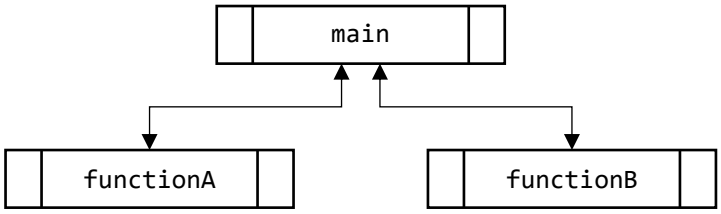
ECE 2305 – Introduction to C Programming
Programming Tutorial 18
User-defined Functions and Function Calls

Write a C++ program to demonstrate user-defined functions and function calls as illustrated below.



The functions are described below in "pseudo code."

functionA (local variable *n*)
    n = *n* + 1
    display the number of times functionA has been called
    display Memory Map for functionA
        variable *n* address and value
    return the value of *n*

functionB (local variable *n*)
    n = *n* + 1
    display the number of times functionB has been called
    display Memory Map for functionB
        variable *n* address and value
    return the value of *n*

main function:
    declare: variable *a* = the number of times functionA is called
    declare: variable *b* = the number of times functionB is called
    menu for main
        display Memory Map for function Main
            variable *a* address and value
            variable *b* address and value
        1. call functionA, send the value of *a*
            assign the return value to *a*
        2. call functionB, send the value of *b*
            assign the return value to *b*
        3. exit

Document the exercise in a Word document.

ECE 2305 – Introduction to C Programming
Programming Tutorial 18
User-defined Functions and Function Calls
Input

```
// ECE 2305 - Tutorial 18
// User-define Functions and Functions Calls
#include<iostream>
using namespace std;
int functionA(int n)
{	n = n + 1;
	cout << "Function A has been called " << n << " times." << endl;
	cout << "Memory Map for Function A" << endl;
	cout << "Name \tAddress \tValue" << endl;
	cout << "n \t" << &n << "\t" << n << endl;
	return n;
}
int functionB(int n)
{
	n = n + 1;
	cout << "Function B has been called " << n << " times." << endl;
	cout << "Memory Map for Function B" << endl;
	cout << "Name \tAddress \tValue" << endl;
	cout << "n \t" << &n << "\t" << n << endl;
	return n;
}
int main()
{
	int a = 0;
	int b = 0;
	int choice = 0;
	do
	{
		system("cls");

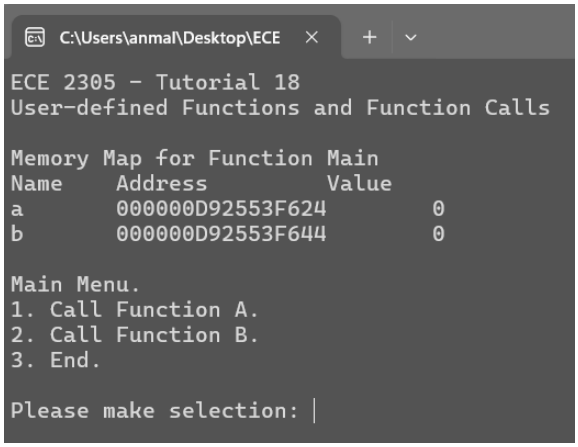
		cout << "ECE 2305 - Tutorial 18" << endl;
		cout << "User-defined Functions and Function Calls" << endl;
		cout << endl;

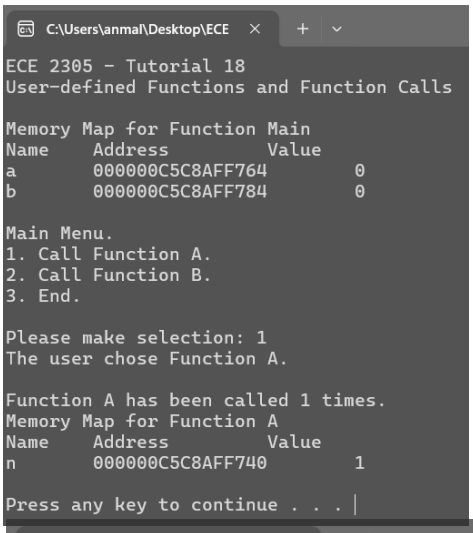
		cout << "Memory Map for Function Main" << endl;
		cout << "Name \tAddress \tValue" << endl;
		cout << "a \t" << &a << "\t" << a << endl;
		cout << "b \t" << &b << "\t" << b << endl;
		cout << endl;

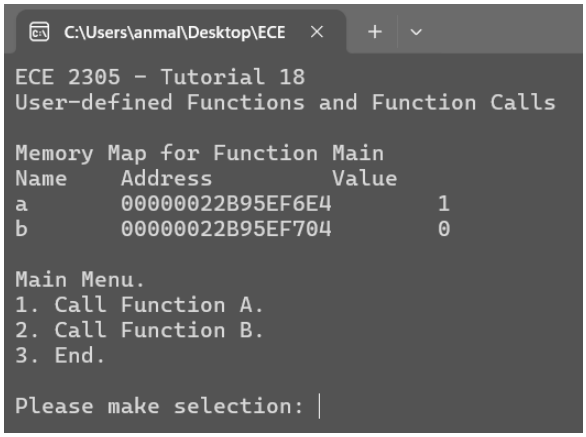
		cout << "Main Menu." << endl;
		cout << "1. Call Function A." << endl;
		cout << "2. Call Function B." << endl;
		cout << "3. End." << endl;
		cout << endl;
		cout << "Please make selection: ";
		cin >> choice;
		switch (choice)
		{
		case 1:
			cout << "The user chose Function A." << endl;
			cout << endl;
			a = functionA(a);
			cout << endl;
			system("pause");
			break;
```

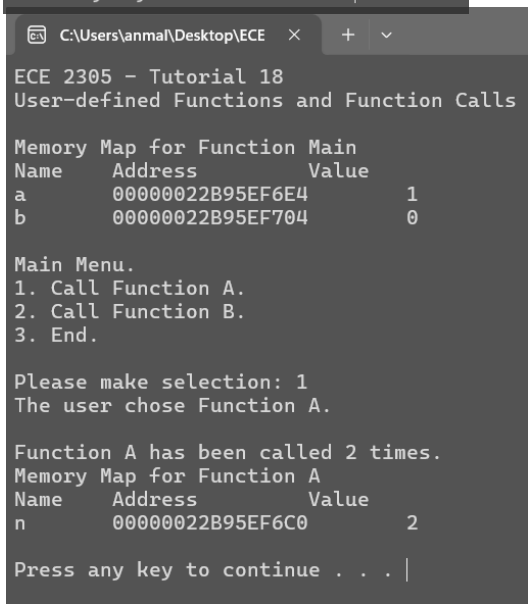
```
                case 2:
                        cout << "The user chose Function B." << endl;
                        cout << endl;
                        b = functionB(b);
                        cout << endl;
                        system("pause");
                        break;
                case 3:
                        cout << "The user chose to end." << endl;
                        cout << endl;
                        system("pause");
                        break;
                default:
                        cout << "Please make a proper choice." << endl;
                        system("pause");
                }
        } while (choice != 3);
        return 0;
}
```

Output









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